State Emergency Management Plan

Radiation Sub-Plan



VICTORIA State Government | Department of Health



To receive this document in another format, email Emergency Management Policy <EM_Policy@health.vic.gov.au>.

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Contents

Abbreviations 4

1 Introduction 5

1.1 Purpose 5

1.2 Objectives 5

1.3 Scope 6

1.4 Implementation 6

1.5 Authorising environment 6

1.6 Audience 7

1.7 Assumptions 8

1.8 Exercising and evaluation 8

1.9 Review 8

1.10 Linkages and hyperlinks 8

2 Context 9

2.1 What is radiation? 9

2.2 Radiation emergencies 9

3 Mitigation and preparedness 10

3.1 Victorian government agencies 10

3.2 Planning 11

3.3 National support 12

4 Public information and community engagement 13

5 Response 14

5.1 Notifications 14

5.2 Escalation of arrangements 14

5.3 Roles and responsibilities 16

5.4 Community protection and information 21

5.5 National support 21

5.6 Consequence management 21

5.7 Relief (as part of response) 22

5.8 Transition to recovery 22

6 Recovery 23

Appendix A: Key supporting information 24

Legislation 24

Plans, arrangements and guidelines 25

Governance forums 26

Appendix B: Glossary 28

Appendix C: Public information and community engagement arrangements 30

Governance 30

Planning 30

Implementation 31

Abbreviations

Abbreviation	Description
ACCHO	Aboriginal community-controlled health organisation
ANSTO	Australian Nuclear Science and Technology Organisation
ARPANSA	Australian Radiation Protection and Nuclear Safety Agency
AV	Ambulance Victoria
CBRNe	chemical, biological, radiological, nuclear and high-yield explosives
CCP	Chief Commissioner of Police
COMDISPLAN	*Australian Government disaster response plan 2020*
CFA	Country Fire Authority
CHO	Chief Health Officer
DFFH	Department of Families, Fairness and Housing
DH	Victorian Government Department of Health
DTP	Department of Transport and Planning
ED	emergency department
EMV	Emergency Management Victoria
EMJPIC	Emergency Management Joint Public Information Committee
EPA	Environment Protection Authority
ERV	Emergency Recovery Victoria
EMC	Emergency Management Commissioner
FAT	Field Assistance Team
FRV	Fire Rescue Victoria
FEMO	Field Emergency Management Officer
HAZMAT	hazardous materials

Abbreviation	Description
HESP	*Health Emergencies Sub-Plan*
IAEA	International Atomic Energy Agency
IC	Incident Controller
IMT	Incident Management Team
JSOP	Joint Standard Operating Procedure
LPHU	Local Public Health Unit
MEMP	municipal emergency management plan
MEMPC	Municipal Emergency Management Planning Committee
NEMA	National Emergency Management Agency
NGO	non-government organisation
NPW	nuclear-powered warship
PHN	Primary Health Network
PPE	personal protective equipment
RelLA	Relief Lead Agency
RelSA	Relief Support Agency
RecLA	Recovery Lead Agency
RecSA	Recovery Support Agency
REMP	regional emergency management plan
REMPC	Regional Emergency Management Planning Committee
SEMP	*State emergency management plan*
SEMT	State Emergency Management Team
SHEMC	State Health Emergency Management Coordinator
SHEMT	State Health Emergency Management Team

1 Introduction

1.1 Purpose

The *State emergency management plan* (SEMP) Radiation Sub-Plan (the plan) outlines the integrated and coordinated whole-of-government approach required to respond effectively to radiation emergencies in Victoria.

Radiation emergencies are emergencies that cause consequences to human life, health, property or the environment from a radioactive source.

Under the SEMP, the Victorian Department of Health (DH) is the control agency for emergencies involving radioactive materials (including leaks and spills). DH is responsible for the maintenance and communication of this plan.

In the event of a radiation emergency occurring concurrently with one or more other emergencies, the State Controller – Health must consult with the Emergency Management Commissioner (EMC) and other relevant control agencies to determine and document appropriate control arrangements.

This plan incorporates feedback from external consultation as required under the *Emergency Management Act 2013*. This includes a broad range of Victorian emergency service organisations and related agencies.

For information regarding emergency management arrangements for other hazards with potential health risks, refer to Emergency Management Victoria's (EMV) website for corresponding SEMP Sub-Plans <https://www.emv.vic.gov.au/responsibilities/state-emergency-management-plan-sub-plans>.

1.2 Objectives

The objectives of this plan are guided by:

- the SEMP, which aims to:
  - provide clarity on roles, responsibilities, escalation and communication channels to enable an effective and efficient response to a radiation emergency
  - set out arrangements to integrate and harmonise multiagency emergency responses to such incidents
  - identifies and applies national and international standards, where applicable.
- the SEMP *Health Emergencies Sub-Plan,* which aims to:
  - maximise health outcomes by providing treatment in a safe, timely and coordinated manner
  - minimise impacts on the health system
  - provide timely, tailored and relevant health information and warnings to the community, to assist community members to make informed decisions about their safety
  - ensure clarity about roles, responsibilities, escalation and communication channels to enable an effective and efficient health emergency response, particularly for the health system.
- the International Atomic Energy Agency's (IAEA) General safety requirements (GSR) Part 7: *Preparedness and response for a nuclear or radiological emergency,* which aims to:
  - save lives and reduce the risk or mitigate the short- and long-term health impacts and consequences of a radiation emergency
  - reduce or mitigate non-radiological consequences
  - protect property and the environment
  - prepare for the resumption of normal social and economic activity and return to normal operations to the extent it is possible and can be done safely.[1]

1 International Atomic Energy Agency (IAEA) 2015, *Preparedness and response for a nuclear or radiological emergency,* <https://www.iaea.org/publications/10905/preparedness-and-response-for-a-nuclear-or-radiological-emergency>.

1.3 Scope

This plan is for radiation emergencies that require a multiagency response.

1.3.1 Out of scope

If the situation involves counterterrorism, or a warlike act, DH will be a support agency. It will work with the Chief Commissioner of Police (CCP). Victoria Police will assume State Control as advised under the SEMP.

This plan does not include the details about operational arrangements or plans that are maintained by individual departments and agencies.

This plan exists parallel to the *Special plan for nuclear-powered warships (NPW) to Melbourne – Victoria*. However, section 5.7: 'Relief' and section 6: 'Recovery' in this plan should be enacted for an NPW event.

1.4 Implementation

The arrangements in this plan apply on a continuing basis and provides strategic information regarding Victoria's mitigation, preparedness, response, and recovery from radiation emergencies.

This plan is intended to interact with other SEMP Sub-Plans and is supported by agency operational plans. This plan will be applied in conjunction with the arrangements set out in the SEMP *Health Emergencies Sub-Plan* (HESP) <https://www.emv.vic.gov.au/responsibilities/state-emergency-management-plan-sub-plans/semp-health-emergencies-sub-plan>.

1.5 Authorising environment

The *Emergency Management Act 1986 and 2013* provides the legislative basis to manage emergencies in Victoria.

The SEMP outlines provisions for mitigation, response and recovery arrangements for emergencies. It specifies the roles and responsibilities of agencies in relation to emergency management.

This plan is a subordinate plan to the SEMP and has been approved by the State Crisis and Resilience Council.

For radiation emergencies, further Victorian legislation that applies to specific powers and responsibilities includes the following.

Radiation Act 2005

The *Radiation Act 2005* provides for the regulation of radiation practices and of individuals who use radiation sources.

Radiation practices include possession of radiation sources including radioactive material, transport of radioactive material and disposal of radiation sources.

Section 93 of the *Radiation Act 2005* provides for the Secretary, Department of Health to authorise the use of emergency powers by Authorised Officers appointed under section 69.

Under the instrument of delegation, the power to authorise the use of emergency powers is delegated to the Chief Health Officer (CHO).

For further information on governance structures, legislation, plans and guidelines relating to radiation, refer to Appendix A: Key supporting information.

Public Health and Wellbeing Act 2008

The *Public Health and Wellbeing Act 2008* provides the legal framework for protecting and promoting public health in Victoria. It establishes powers and responsibilities for the CHO and the Secretary including the delegation of powers.

National and international standards

Peak national and international bodies which develop standards, codes and guides that provide advice to support radiation protection in Victoria include, but are not limited to:

- Australian Radiation Protection and Nuclear Safety Agency (ARPANSA)
- Australian Nuclear Science and Technology Organisation (ANSTO)
- International Atomic Energy Agency (IAEA).

1.6 Audience

The audience for this plan comprises all relevant agencies and healthcare providers. In particular, this includes the emergency management sector and health sector with a role in the preparedness for, response to and recovery from a radiation emergency.

This includes the Victorian Government and agencies, local government, emergency services, health services, regulated entities and businesses.

Community members may find the contents of this plan informative to prepare for, respond to and recover from radiation emergencies.



1.7 Assumptions

This plan is based on the following assumptions:

- The reader is familiar with the SEMP, which outlines the holistic details of the arrangements for an integrated, coordinated, and comprehensive approach to emergency management in Victoria.
- The reader is familiar with the HESP, which outlines the coordinated arrangements for managing the risks and consequences of health emergencies in Victoria, including pandemics.
- The reader is familiar with the *Victorian emergency operations handbook* <https://www.emv.vic.gov.au/publications/victorian-emergency-operations-handbook>, which provides a reference guide to the key mechanisms (structures, systems, roles and functions) required to effectively and safely deliver incident management operations for major emergencies in Victoria.
- Agency plans and supporting documents provide the details of relevant activities of individual agencies.

1.8 Exercising and evaluation

DH will exercise this plan within one year of the published date. The exercise will be evaluated and, where improvements are required, the plan will be amended and a revised version issued.

Exercises will be conducted in accordance with the Australian Institute for Disaster Resilience (AIDR) *Managing exercises handbook* <https://knowledge.aidr.org.au/resources/handbook-managing-exercises> and in conjunction with the ARPANSA *Guide for radiation protection in emergency exposure situations* (2019) <https://www.arpansa.gov.au/regulation-and-licensing/regulatory-publications/radiation-protection-series/guides-and-recommendations/rpsg-3>.

The department will evaluate the exercise through a debrief process after this plan is activated and response activities cease.

All participating agencies, including recovery agencies, will be represented to evaluate response arrangements and recommend improvement opportunities for future events. Evaluations should consider the *Lessons management framework* (EM-LEARN) <https://www.emv.vic.gov.au/how-we-help/reviews-and-lessons-management/lessons-management-framework-em-learn> and AIDR *Lessons management handbook* <https://knowledge.aidr.org.au/resources/lessons-management-handbook/>.

1.9 Review

This plan was current at the time of publication and remains in effect until modified, superseded or withdrawn.

DH will review and update this plan every three years. More frequent reviews may be undertaken if required, for example following an event, exercising of the plan, or following substantial change to relevant legislation or machinery of government arrangements.

1.10 Linkages and hyperlinks

This plan does not duplicate information provided elsewhere. Instead, it links to resources and websites. These are accurate at the time of publication.

Resources identified frequently within the plan (for example, SEMP and HESP) are not hyperlinked in each instance.

Appendix A: Key supporting information has links to relevant state, national and international governance structures, legislation and plans related to this plan.

2 Context

2.1 What is radiation?

Australians are exposed to radiation from natural and artificial sources every day.

There are two main types of radiation: non-ionising and ionising. They are categorised according to how much damage the energy or wavelength of the source can affect our health.

Non-ionising radiation does not have enough energy to damage DNA directly. Examples include ultraviolet (UV) light, powerlines and electrical equipment.

Ionising radiation is radiation that passes through the body and causes damage to DNA. Examples are alpha particles, gamma rays, X-rays and neutrons.[2]

The IAEA's 'What is radiation' webpage <https://www.iaea.org/newscenter/news/what-is-radiation> has more information about radiation.

2.2 Radiation emergencies

Although the likelihood of a radiation emergency is relatively low, the consequences could be extremely high. A radiation emergency would cause considerable fear and anxiety within the community.

Examples of potential radiation emergencies in Victoria include:

- public exposure and contamination from unknown origins
- loss of control of a high-activity radiation source
- terrorist threats or acts (note Victoria Police is the control agency for this type of event)
- transport accidents (including rail, road, water)
- an accident involving an irradiation facility (such as industrial irradiators)
- a significant release of radioactive material from a storage facility.

Agencies responding to a radiation emergency should:

- maintain responders' awareness of their role in managing radiation emergencies
- use specific radiation detection equipment to evaluate the risk from radiation sources (for example, human senses cannot detect ionising radiation)
- remember that the health effects of radiation exposure can take days, weeks or years to manifest
- use careful risk communication, because radiation is widely perceived as highly dangerous.

2 International Atomic Energy Agency (IAEA) 'What is radiation' webpage <https://www.iaea.org/newscenter/news/what-is-radiation>.

3 Mitigation and preparedness

Mitigation and preparedness seek to reduce the severity of an emergency and adverse consequences to communities.

The government mitigates the potential for radiation emergencies by regulating radiation practice, licensing users of radiation equipment and monitoring compliance.

Diagnostic, therapeutic, industrial and other uses of radiation have contributed to the safety and quality of life for all Victorians. However, the use of radiation does involve hazards if it is used inappropriately or unnecessarily.

DH regulates the use of radiation to protect people and the environment from its harmful effects. It does this by licensing users of radiation sources and managers of radiation practices under the *Radiation Act 2005*.

DH 'Radiation' webpage <https://www.health.vic.gov.au/public-health/radiation> has more information.

Government agencies, emergency service organisations and health care providers mitigate risks by developing and exercising local arrangements, learning from domestic and international events or shared experiences.

Communities and individuals are supported by a range of activities which include seamless and coordinated public information tools, accessible platforms and services and coordinated messaging to all Victorians.

3.1 Victorian government agencies

Victorian government departments and agencies should identify their role in preparing for a radiation emergency.

They should ensure their plans:

- define their operational arrangements including radiation response capabilities and capacity, education and training
- outline any responsibilities for providing specialist resources and monitoring equipment (such as HAZMAT technicians, scientific analysis and interpretation, radiation detection and expertise) where applicable
- describe Occupational Health and Safety arrangements including supply and use of personal protective equipment (PPE).

See Table 8 of the SEMP <https://www.emv.vic.gov.au/responsibilities/semp/roles-and-responsibilities/mitigation> for a list of mitigation activity undertaken by agencies based on significant risks by emergency.

For resources regarding risks to public health and the health system in Victoria, refer to HESP section 4: 'Mitigation and preparedness'.

As control agency for preparing for a radiation emergency, DH is responsible for:

- exercising this plan
- engaging with community, state and local government and key partners in the health sector to prepare for radiation emergencies
- developing and implementing public health plans, guidelines and advice that raise awareness about health risks and how to mitigate the risk.

Additional responsibilities for DH as control agency refer to HESP chapter 3.1 'Department of Health as the control agency'.

3.2 Planning

There is a shared responsibility for all Victorians to plan for emergencies. Planning activities form part of mitigation and preparedness. They include developing, exercising and reviewing plans.

All emergency management plans should support the needs and concerns of people most at risk.

These plans should also contain provisions for engagement with Aboriginal community-controlled health organisations and Traditional Owners.

3.2.1 State planning

The SEMP provides for an integrated, coordinated and comprehensive approach to plan for, respond to and recover from all emergencies. The EMC is responsible for the preparation of the SEMP.

DH, as the control agency for emergencies involving radioactive materials (including leaks and spills), is responsible for the maintenance and communication of this plan.

3.2.2 Regional planning

At the regional tier, Regional Emergency Management Planning Committees (REMPCs) are responsible for regional emergency management planning, including the development of regional emergency management plans (REMPs).

REMPs should identify how the region will prepare for, respond to, and recover from all hazards.

REMPs should align with state planning to ensure activities and arrangements are coordinated and integrated; leveraging local resources and community networks.

3.2.3 Municipal planning

Under the *Emergency Management Act 2013,* local governments chair multiagency municipal emergency management planning committees (MEMPCs).

MEMPCs assess municipal-level risks and plan accordingly. They are responsible for preparing municipal emergency management plans (MEMPs).

MEMPs outline the agreed municipal-level emergency management arrangements by all participating emergency services, government and non-government organisations (NGOs) in the municipal footprint.

The role of local government and other local agencies documented in MEMPs or MEMP subplans for radiation emergencies must be consistent with state plans, including the SEMP and this plan.

3.3 National support

The Australian Government establishes support arrangements to assist with national health emergencies that impact Victoria.

For more information on the role of the Australian Government for health emergencies, refer to the HESP, section 4.2 'Australian Government'.

The following sections set out support arrangements from the Australian Government specific to radiation emergencies.

3.3.1 Non-financial disaster support

The Victorian Government can request additional resources to assist disaster response and recovery from the Australian Government. These arrangements set out in the *Australian Government disaster response plan* (COMDISPLAN) <https://www.homeaffairs.gov.au/emergency/files/plan-disaster-response.pdf>.

The EMC or CCP, as the Victorian nominated officials, will make a request for non-financial assistance to the National Emergency Management Agency (NEMA) for consideration by the Deputy Coordinator General Emergency Management and Response.

COMDISPLAN requests for physical assistance may include:

- specialist technical advice and expertise
- additional resources
- specialist capabilities.

Refer to 'Appendix A: Key supporting information' for further information on Commonwealth agencies.

3.3.2 National Medical Stockpile

The Australian Department of Health and Aged Care maintains a stockpile of treatments to be mobilised to aid in the management of radiological events.

The CHO or Deputy Chief Health Officers will approve the request to be submitted to the National Medical Stockpile in the event of a radiation emergency requiring additional resources.

4 Public information and community engagement

An emergency involving radiation is likely to cause fear and anxiety within the community. Communications should be accurate, accessible, timely, tailored and clearly explain how people can stay safe.

Messaging will need to consider:

- those who may have been exposed to radioactive materials and need to be decontaminated
- those who may need to shelter in place
- those who may need to evacuate.

Comprehensive, far-reaching, and ongoing communications and engagement approaches, methods and channels may be required to support affected individuals and communities.

It is essential that Victorian Government departments and agencies provide seamless and coordinated public information and community engagement at every stage of managing a radiation emergency.

Refer to 'Appendix C: Public information and community engagement' for more information.



5 Response

This plan is supported by agency operational response plans, protocols and guidelines that provide additional detail to meet their responsibilities during a radiation emergency.

Together, the doctrine facilitates both a collaborative and scalable approach to emergency response, and continual assessment to best meet response needs.

The level of human resourcing, including the number of people and types of skills and expertise required, depends on the type of emergency, the complexity and the impact of the event or incident.

Guides, protocols and further information about Victoria's state health emergency response arrangements can be found on DH's website <https://www.health.vic.gov.au/emergencies/state-health-emergency-response-arrangements>.

Actions required for radiation emergencies include:

- communication of timely, relevant and tailored warnings and information to the community
- consideration of evacuation or shelter in place or indoors
- management of the immediate needs of affected people and the impacts upon the community
- integration of relief activities with response activities
- ongoing assessment and management of the impact and consequences of the emergency
- ensuring that control measures are in place at appropriate levels and an effective incident control structure is in place to achieve protection of life and property
- effectively led, managed and engaged Emergency Management Teams at state, region and incident levels
- longer-term actions, such as the monitoring and control of foodstuffs and relocation and resettlement of affected people.

Radiological emergencies can have wide consequences. Although the initial priority is to protect public health from exposure to radiation, other consequences also need to be addressed in a timely manner. For example, radiological emergencies can have adverse non-radiological consequences such as economic, social and psychological impacts that also require management to ensure they are minimised.[3]

5.1 Notifications

5.1.1 Notification to DH

The initial notification of a radiation emergency can come from several sources. These may include emergency response agencies, government agencies, health services, laboratories and the public.

All agencies alerted of a radiation incident should immediately notify DH through 24/7 Telephone Messaging Centre on **1300 790 733**.

5.1.2 Notification from DH

Key stakeholders will be notified of radiation emergencies via state emergency management arrangements.

5.2 Escalation of arrangements

In line with the SEMP, Victoria has three operational tiers (incident, region and state) <https://www.emv.vic.gov.au/responsibilities/state-emergency-management-plan-semp>. There is the option for an 'area of operation' to be declared to manage a complex emergency that may be geographically located over several municipalities or several regions.

It is highly likely that a radiation emergency will be initially managed at the state tier, noting there will be local incident management required at the scene.

3 IAEA 2020, *Considerations in the development of a protection strategy for a nuclear or radiological emergency,* <https://www-pub.iaea.org/MTCD/Publications/PDF/EPR-Protection_Strategy_web.pdf>.

5.2.1 Incident tier (management at the scene)

Incident control arrangements

For most incidents it is likely that an emergency response agency such as Fire Rescue Victoria (FRV), CFA or Victoria Police will be first on scene and assume the role of Incident Controller (IC).

If radioactive material is suspected to be involved, DH on-call Radiation Officer (1300 790 733) must be immediately contacted to undertake a radiation assessment.

Transfer of incident control may occur when:

- A DH Radiation Field Assistance Team (FAT) undertakes an assessment and determines there is a radiation emergency. Assessments may be conducted virtually or require on-site visit
- it is agreed by the on-site response agency IC and the DH State Agency Commander that transition of control is required
- The DH State Agency Commander will initiate the process for formal appointment of a State Controller – Health and DH IC.
- DH State Agency Commander will undertake the activities of IC until formal appointment can be made.

Contain site and scene safety

The first responding agency IC will conduct an assessment to identify risk as prescribed in agency standard operating procedures.

Based on the initial assessment, and in consultation with the DH Radiation FAT the following should be established:

- safety for first responders
- a **safety perimeter**, which is the boundary of the inner cordoned area (around a dangerous radioactive source)
- a **security perimeter**, which is the boundary of the outer cordoned area
- monitoring requirements, noting not all agencies are equipped with radiation monitoring equipment.

Evacuation or shelter in place and/or indoors

The scale of **evacuation** and/or **shelter in place and/or indoors** will be recommended by the IC with advice from the DH Radiation FAT.

If it is determined evacuation is required, Victoria Police will provide assistance with cordon and containment and road authorities will assist with traffic management.

All evacuation and shelter in place arrangements should be undertaken in line with appropriate JSOPs.

Traffic management

The IC will consult with Victoria Police to establish arrangements for control of the movement of traffic through an incident area.

Traffic management should, where applicable, include management of public transport, freight and potentially ports dependent on location.

Traffic management arrangements should be undertaken in line with appropriate JSOPs.

5.2.2 Regional tier

It is highly likely a radiation emergency will be initially managed at the state and incident tiers. However, regional emergency management and healthcare providers should still identify how they will prepare for, respond to and recover from a radiation emergency.

Regional arrangements should align with state planning to ensure activities and arrangements are coordinated and integrated, leveraging local resources and community networks.

A region, if established, may be an emergency management region as identified in the SEMP, a health region or a region determined at the time based on the geographic area affected by the health emergency.

The regional tier, if established, will provide the first level of assurance and control, coordination of response, recovery, consequence management, communications, and resourcing. The Regional Controller will report to the State Controller – Health.

A Health Emergency Management Liaison Officer will be appointed and work within the Regional Incident Control Centre.

When a response at the regional tier is required and appropriate for the effective coordination of an incident, the Regional Health Coordinator will coordinate emergency response activities across supporting agencies and departments and the health system (including hospitals, primary health and other acute services).

The Regional Health Coordinator, where appointed, will report to the State Health Coordinator, who ultimately reports to the State Health Emergency Management Coordinator.

5.2.3 State tier

In addition to arrangements outlined as per SEMP and HESP, a dedicated Public Health Advice Cell will be formed to provide radiation subject matter expertise into the Intelligence section of the State Control Team.

5.3 Roles and responsibilities

The SEMP provides agency roles and responsibilities for responding to emergencies in Table 9 (control agencies for response) and Table 10 (support agencies for response) <https://www.emv.vic.gov.au/responsibilities/semp/roles-and-responsibilities/response>. For State Control arrangements, refer to SEMP and HESP.

The roles and responsibilities identified in this plan are not exhaustive and should be considered in conjunction with relevant legislation, plans, frameworks and guidelines.



5.3.1 Department of Health

DH is responsible for:

- discharging its responsibilities under the arrangements in the SEMP as Control Agency for radioactive materials accidents (including leaks and spills)
- providing a State Controller to oversee the control arrangements, consequence management planning and integration of recovery with the response arrangements
- providing an IC to oversee cross-agency operations during a radiation emergency
- ensuring public information and messaging is coordinated and provides timely, accurate public information
- providing radiation expertise such as a Senior Radiation Officer to support operational responses including mobile monitoring and decontamination as required
- ensuring arrangements are in place for the health sector to effectively prevent, prepare for, respond to, and recover from radiation emergencies and mitigate adverse health consequences for the community
- leading and establishing integrated arrangements with government departments, agencies and sectors to support emergency arrangements
- monitoring, detecting and investigating radiation emergencies as soon as practicable
- restoring radiation safety and minimising radiation exposure as far as practicable
- identifying and containing the radiation source
- assisting FRV and the Country Fire Authority (CFA) with decontamination procedures
- providing assistance and advice for site clean-up and coordination of safe waste disposal methods and pathways
- sourcing additional capability, such as specialist radiation expertise
- managing health consequences of the radiation emergency to support an effective response and recovery, including:
  - reducing the risk of severe **deterministic** health effects in workers and the public (high doses leading to skin burns, tissue damage or death)
  - reducing the risk of **stochastic** health effects in the population (low doses over long periods leads to increased cancer risk)
- ensuring the health system can effectively respond and mitigate the adverse consequences for communities
- advising the Radiation Emergency Coordination Centre at ARPANSA of a radiation emergency.

For DH Role Statement under the SEMP, see EMV's webpage Department of Health – Role statement <https://www.emv.vic.gov.au/responsibilities/state-emergency-management-plan-semp/roles-and-responsibilities/role-statements/department-of-health-role-statement>.

5.3.2 Key support agencies

Any government or non-government agency may be requested to assist DH in a radiation emergency response, including relief.

Support agencies should have arrangements in place to fulfil their responsibilities as outlined in SEMP role statements and consider what may be required of them to support emergency management arrangements for radiation emergencies.

This includes responding agencies providing their own responders with appropriate PPE in keeping with occupational workplace and safety obligations.

The safety and the protection of human life, including emergency services personnel and community members, takes priority above all other considerations for managing radiation emergencies. This is in line with the 'State emergency management priorities' <https://www.emv.vic.gov.au/responsibilities/state-emergency-management-priorities>.

Ambulance Victoria

Time-critical patients may be transported prior to decontamination where they are at greater clinical risk.

Ambulance Victoria will:

- triage patients and determine treatment priority
- provide pre-hospital clinical care
- transport and distribute patients to appropriate medical care
- provide health support to patients undergoing decontamination, where possible
- provide health support to other agencies, where appropriate
- deploy a Health Commander to direct the operational health response and assemble and lead the Health Incident Management Team
- represent Health (pre-hospital) as a member of the State Health Emergency Management Team (SHEMT)
- activate other key position holders or mobile specialist teams, including activation of the Field Emergency Medical Officer Program, Victorian Medical Assistance Teams and first aid agencies, if required
- initially notify the hospitals receiving patients
- work with DH to provide relevant health information and advice to support broader community health inquires and availability of alternative service providers based on case definition.

The HESP provides general principles for managing mass casualties and subordinate State Health Emergency Arrangements, including *Mass casualty and pre-hospital operational response plan (incident tier)* <https://www.health.vic.gov.au/publications/mass-casualty-and-pre-hospital-operational-response-plan-incident-tier>.

Fire Rescue Victoria

Fire Rescue Victoria will:

- ensure DH is notified
- perform an incident size up
- obtain information on the incident, casualties and circumstance(s)
- deploy radiation detection equipment.
- set-up decontamination with advice from radiation specialist(s)
- perform rescue under advice from radiation specialist(s)
- ensure appropriate PPE will be deployed to manage decontamination operations/ procedures.

Country Fire Authority

Country Fire Authority will:

- ensure DH is notified
- provide decontamination and equipment at scene.
- implement decontamination procedures for contaminated and exposed or potentially exposed persons, under DH direction.
- ensure appropriate PPE will be deployed to manage decontamination operations/ procedures.

Victoria Police

Victoria Police will:

- ensure all routine and emergency communications and operational requirements are met
- undertake evacuation in consultation with DH and road authorities and in line with corresponding Joint Standard Operating Procedure (JSOP).

Healthcare providers

Hospitals

DH in collaboration with health services, will support the coordination of impacts on the health system, including managing healthcare expertise and resources.

It is expected that a health service has arrangements in place to respond to and manage emergencies where decontamination of patients may be required on site.

Hospitals will work closely with the DH Emergency Management State Operations, initially through the State Duty Officer – **1300 790 733**.

Primary Health Network

Primary Health Networks (PHN) will:

- coordinate access to primary care resources, including general practitioners (GPs), pharmacy, community mental health and alcohol and drug services for emergency response
- facilitate the delivery of warnings and public health messages to the primary care sector
- commission health services to support community recovery.

Aboriginal health

Engagement with Aboriginal communities and Aboriginal community-controlled organisations will promote Aboriginal community self-determined priorities to be addressed throughout the management of radiation emergencies in a culturally safe manner.



Department of Energy, Environment and Climate Action

Agriculture Victoria

Agriculture Victoria will:

- provide advice to DH on agricultural produce that may have been impacted (with regard to primary production and primary processing)
- provide advice to affected agriculture sectors, including regulators, about risks and risk mitigation measures, relief and recovery
- provide an Emergency Management Liaison Officer, if required, to support DH.

Wildlife

Wildlife will:

- be responsible for the provision of advice on any wildlife welfare implications arising from the incident
- undertake stakeholder engagement with the wildlife welfare sector, with DH-approved messages
- provide an Emergency Management Liaison Officer, if required, to support DH.

Environment Protection Authority

The Environment Protection Authority (EPA) will provide advice on environmental matters, where required.

The Chief Environmental Scientist (CES) has an important role during emergencies involving significant pollution consequences by supporting the EMC, State Controller and Chief Health Officer with expert advice on the practical measures to protect the environment and public health from the impacts of pollution and waste. The CES also provides a trusted and authoritative public face for the communication of complex pollution and waste advice.

The powers of the *Radiation Act 2005* extend to the protection of the environment.

Local Public Health Units

Local Public Health Units (LPHUs) have a role to play in providing locally based support to DH in all emergencies including radiation emergencies.

LPHUs can assist in disseminating public health messaging and support engagement with local communities, collaborate with community partners, GPs, and hospital-based services to minimise public health risks to the Victorian community.

LPHUs should be aware of local public health risks and hazards and appropriately prepare for these in collaboration with local emergency management arrangements. This will assist in providing seamless and coordinated public information and community engagement response.

Pharmacies

In the event of a radiation emergency, community pharmacies will need to address the challenges of providing pharmaceutical services. Pharmacies located within the affected area will likely cease operation until the area is deemed safe to return. The temporary closure of a pharmacy will result in the discontinuation of pharmaceutical services provided to that community.

Pharmacies in surrounding areas and beyond will begin to service displaced patients without their medications or prescriptions. Due to the nature of the emergency, the usual healthcare organisations that manage patient medication are unlikely to be contactable to verify medication history. DH will need to provide special authorisation to community pharmacists to continue supply of regular medication without prescriptions in this scenario.

Emergency services will need to coordinate closely with pharmaceutical wholesalers to ensure the timely delivery of medication to the pharmacy.

Communications between community pharmacies and their local Primary Health Network (PHN) need to be well established before emergencies arise. Community pharmacies need to be familiar with the method of communication expected from the PHN and provide advice to pharmacies directly affected by the emergency. Pharmacies that continue to provide services will need to collaborate closely with PHNs to refer patients where appropriate.

Department of Transport and Planning

The Department of Transport and Planning (DTP) will assist with the management of road links during emergencies, including route selection and road egress, emergency traffic management, escorting, route conditions advice and control.

DTP will assist Victoria Police establish traffic management points and ports depending on location/source of radiation. This would be done by the Traffic Management Manager, in consultation with DH and road authorities including public transport, freight and potentially ports depending on location/source of radiation.

Traffic management arrangements are further outlined in the corresponding JSOP.

5.4 Community protection and information

Effective communication with the public and other stakeholders during a radiological emergency supports actions to protect health.

Communication needs to be transparent, timely, clear, factually correct, objective and easily understood.

Use targeted communications and engagement channels to reach specific population groups. These may include Aboriginal and Torres Strait Islander people, those from multifaith and multicultural communities, people with vision impairment, and those who are deaf or hard of hearing.

Effective public communication will:

- encourage people at risk to protect themselves
- reassure people who are not at risk that they are safe
- help to minimise rumours and fears
- facilitate relief efforts and help maintain public trust and confidence in the government's response.

Refer to 'Appendix C: Public information and community engagement arrangements' for more information.

5.5 National support

If additional radiation response resources and assistance is required, these will be requested through the COMDISPLAN.

The EMC will make a request for non-financial assistance to the NEMA for consideration by the Deputy Coordinator General Emergency Management and Response.

5.6 Consequence management

Under the *Emergency Management Act 2013,* the EMC is responsible for consequence management for major emergencies.

The State Controller – Health will support the EMC to ensure that the consequences of potential or actual impacts are managed. Refer to HESP section 8 'Consequence management' for further information.

DH will work with government departments and agencies, the emergency management sector, the health sector, industry, businesses and the community to identify and mitigate potential consequences of a radiation emergency.

5.7 Relief (as part of response)

Relief is the provision of assistance to meet the essential needs of individuals, families and communities during and in the immediate aftermath of an emergency.

Relief activities should:

- begin as soon as necessary, which is often during the response phase
- be adaptable, depending on the scale and complexity of the radiation incident or emergency
- recognise and consider communities' diversity
- continue into the immediate aftermath of a radiation emergency.

The severity and scale of the radiation emergency will determine relief considerations required for affected communities. These will likely include the establishment of relief centres if mass relocation or evacuation is required, or shelter in place arrangements.

The delivery of relief services is coordinated and managed by Relief Lead Agencies (RelLAs) and supported by Relief Support Agencies (RelSAs), as identified in the SEMP.

The scale and complexity of a radiation emergency will determine which tiers and relief services will need to be activated.

DH will provide expertise to the State Relief and Recovery Team to support relief and recovery arrangements for radiation emergencies (for example, monitoring and control of foodstuffs, water quality, relocation and resettlement of affected people, decontamination of contaminated environment, psychosocial impacts, long-term health impacts).

5.8 Transition to recovery

The scale and complexity of a radiation emergency will determine transition requirements to support the coordination of recovery arrangements.

The State Controller – Health is responsible for:

- leading and facilitating the transition to recovery
- notifying relevant emergency service organisations, public health services, private hospitals, the primary health sector and other health sector stakeholders of the stand down of response activities
- ensuring an operational debrief of all participants to learn from the emergency management experience is conducted
- considering support, advice and information for personnel involved in a response
- returning to 'business as usual' or the 'new normal'.

Collaboration between response and recovery agencies is required for a seamless transition to recovery. All agencies involved should cooperatively share information, planning and decision making to execute plans.

Once the emergency response activities have concluded and where relief and recovery activities need to continue, the arrangements for managing the emergency will formally transition to recovery.

6 Recovery

Emergency management in Victoria positions local government as the lead agency responsible for recovery at the local level.

In events where complexity and scale require regional or state coordination of recovery arrangements, this is facilitated at through Emergency Recovery Victoria (ERV).

Following an emergency, recovery coordination and supports are organised around five lines of recovery to achieve long-term outcomes:

- people and wellbeing
- Aboriginal culture and healing
- biodiversity and environment
- business and economy
- buildings and infrastructure.

ERV acts to ensure that planning for recovery is integrated and that community voices and needs are kept at the centre of recovery planning and coordination.

Recovery activities should commence immediately and be informed by impact assessment data to reduce the severity of impacts from a radiation emergency. This will ensure minimal impacts to short and long-term health, environment, disruption to essential services and critical infrastructure.

DH will provide expertise to the State Relief and Recovery Team to support relief and recovery arrangements for radiation emergencies (for example, monitoring and control of foodstuffs, water quality, relocation and resettlement of affected people, decontamination of contaminated environment).

The Recovery Lead Agency (RecLA) will lead and coordinate recovery arrangements to ensure the needs of individuals and communities are met with support from Recovery Support Agencies (RecSAs).

Victoria's recovery arrangements align with the *National principles for disaster recovery* <https://knowledge.aidr.org.au/resources/national-principles-for-disaster-recovery/>.

The SEMP Tables 13 to 18 provide roles and responsibilities for the coordination of recovery activities and services <https://www.emv.vic.gov.au/responsibilities/semp/roles-and-responsibilities/recovery-co-ordination>.

Refer to SEMP and HESP for further information regarding recovery arrangements.

Appendix A:
Key supporting information

The following tables provide relevant state, national and international legislation, plans, and key governance forums related to this plan. This list is not exhaustive.

Legislation

Victoria

Legislation	URL
Emergency Management Act 1986	https://www.legislation.vic.gov.au/in-force/acts/emergency-management-act-1986/051
Emergency Management Act 2013	https://www.legislation.vic.gov.au/in-force/acts/emergency-management-act-2013/020
Public Health and Wellbeing Act 2008	https://www.legislation.vic.gov.au/in-force/acts/public-health-and-wellbeing-act-2008/060
Radiation Act 2005	https://www.legislation.vic.gov.au/in-force/acts/radiation-act-2005/033
Radiation Regulations 2017	https://www.legislation.vic.gov.au/in-force/statutory-rules/radiation-regulations-2017/001



Plans, arrangements and guidelines

International

Title	URL
International Atomic Energy Agency (IAEA) Safety Standards series no. GSR Part 7 *General safety requirements*	https://www.iaea.org/publications/10905/preparedness-and-response-for-a-nuclear-or-radiological-emergency

National

Title	URL
Australian Government disaster response plan 2020 (COMDISPLAN)	https://www.homeaffairs.gov.au/emergency/files/plan-disaster-response.pdf
Australian Radiation Protection and Nuclear Safety Agency (ARPANSA) guidelines	https://www.arpansa.gov.au/regulation-and-licensing/licensing/information-for-licence-holders/regulatory-guides
Australian Institute for Disaster Resilience (AIDR) Managing exercises handbook	https://knowledge.aidr.org.au/resources/handbook-managing-exercises/
Australian clinical guidelines for radiological emergencies	https://www.health.gov.au/resources/publications/australian-clinical-guidelines-for-radiological-emergencies?language=en

State

Title	URL
Victorian emergency operations handbook	https://www.emv.vic.gov.au/publications/victorian-emergency-operations-handbook
State emergency management plan (SEMP)	https://www.emv.vic.gov.au/responsibilities/semp
SEMP *Health Emergencies Sub-Plan* (HESP)	https://www.emv.vic.gov.au/responsibilities/semp-sub-plans/semp-health-emergencies-sub-plan

Governance forums

International

Title	Description
International Atomic Energy Agency (IAEA)	The IAEA is an intergovernmental organisation that seeks to promote the peaceful use of nuclear energy and to inhibit its use for any military purpose, including nuclear weapons.

National

Title	Description
Australian Radiation Protection and Nuclear Safety Agency (ARPANSA)	ARPANSA is an Australian Government agency governed under the *Australian Radiation Protection and Nuclear Safety Act 1998* (ARPANS Act). Under the ARPANS Act, ARPANSA is responsible for protecting the health and safety of people, and the environment, from the harmful effects of radiation arising from Commonwealth regulated radiation practices. Assistance may be requested by Victoria to Commonwealth agencies, including ARPANSA, through activation of the COMDISPLAN. The coordination of the Commonwealth emergency response is the responsibility of the NEMA. Referral to ARPANSA is required to follow normal procedures for requests to Commonwealth agencies set out in the SEMP. ARPANSA: • develops codes, standards, guides and provides advice to support radiation protection and nuclear safety throughout Australia • conducts its own internal exercises to ensure adequate preparedness for a radiological or nuclear emergency • facilitates exercises related to nuclear-powered warships through its involvement in the interdepartmental committee, Visiting Ships Panel (Nuclear). ARPANSA is not resourced to coordinate or conduct broader multijurisdictional exercises. No coordinated program exists to involve all relevant roles from first responder through to senior policy and decision makers in exercises.
Australian Nuclear Science and Technology Organisation (ANSTO)	ANSTO is a public research organisation responsible for delivering specialised radiological and nuclear advice, scientific services and products. For prevention, ANSTO partners with the United States National Nuclear Security Administration and the IAEA to form the South East Asian Regional Radiological Security Partnership to improve the security of radioactive sources in South East Asia. For preparedness, ANSTO provides radiation safety training courses for emergency service specialists, such as crime scene and bomb response. ANSTO also conducts research into management of contaminated forensic evidence, and nuclear material characterisation. ANSTO can also assist with vulnerability assessments, technical advice and operational support, including laboratory analysis.

National (continued)

Title	Description
Australian Department of Health and Aged Care	The Australian Department of Health and Aged Care provides overall national coordination of the health response, including liaising with the international community, providing logistics support to states and territories, activating the National Incident Room, and leadership in the coordination of national emergency media management arrangements. In a large-scale emergency involving radiation cases, it is likely that the Australian Government will form an Inter-Departmental Committee (IDC) or taskforce to coordinate the work of Australian Government departments and agencies. The lead agency of the IDC or taskforce will be determined at the time of the emergency.
Australian Defence Force (ADF)	The ADF maintains the Special Operations Engineering Regiment, which may be deployed to assess and respond to radiation emergencies. The circumstances of a terrorist event will determine whether security agencies will declare it a national terrorist situation. If a national terrorist situation is declared, overall responsibility for policy and strategy transfers to the Australian Government, in close consultation with relevant states and territories. This may involve determining overall policy objectives, setting priorities between policy objectives where resources are inadequate, pre-positioning resources, international liaison and determining public communication messages. The Australian Government's role does not include operational management and deployment of emergency services. The Commissioners of Police, including the Commissioner of the Australian Federal Police, will determine the command and resourcing of the national police response. In other respects, the management arrangements in a national terrorist situation will replicate those in other terrorist situations.
National Medical Stockpile	The National Medical Stockpile reserves drugs, vaccines, antidotes and personal protective equipment for use in national health emergencies.

State

Title	Description
Radiation Advisory Committee	https://www.health.vic.gov.au/radiation/radiation-advisory-committee
State Crisis and Resilience Council	https://www.emv.vic.gov.au/about-us/what-we-do/our-legislation

Appendix B: Glossary

Term	Definition
Acute care	Includes admitted and non-admitted services such as critical care, surgical services, Hospital in the Home, specialist clinics, trauma and emergency services.
Casualty	A person who is sick, injured or killed in an emergency.
Deterministic effect (short-term health effects)	Injury of tissues and organs due to cell death. Also termed 'tissue reaction'. Definition derived from the International Commission on Radiological Protection (ICPR) <http://icrpaedia.org/ICRP_Glossary>
Health emergency	Health emergency in the context of this plan includes an incident or emerging risk to the health of community members, from whatever cause, and requires a significant and coordinated effort to ensure the health system can effectively respond and mitigate the adverse health consequences for communities.
Health service	Relates to public health services, denominational hospitals, metropolitan hospitals and public hospitals, as defined by the Health Services Act 1988, with respect to acute and subacute services provided within a hospital or a hospital-equivalent setting.
Ionising radiation	Radiation consisting of particles, X-rays, or gamma rays with sufficient energy to cause ionisation in the medium through which it passes.
Non-ionising radiation	Radiation with insufficient energy to cause ionisation in the medium through which it passes.
People most at risk	Individuals and communities who have the potential to be adversely affected by a disaster or emergency and who, because of the circumstances in their everyday lives, require significant and coordinated priority intervention, response, and support from a variety of government and NGOs and the broader community for their safety.
Pre-hospital	A functional component of health emergency response, from response at the scene of an incident, to the receiving hospital or other healthcare facility.
Primary health	The care received at the first point of contact with the healthcare system, for example, when someone sees a physiotherapist because they have a sore back. It is traditionally delivered by GPs, in community health services and or through private allied health providers.
Public health	Public health involves preventing the occurrence and spread of disease and illness, and reducing the risk posed by potentially dangerous substances to ensure safe environments across Victoria.

Term	Definition
Safety perimeter	A boundary of the inner cordoned area (around a dangerous radioactive source).
Security perimeter	A boundary of the outer cordoned area.
Shelter in place	The process by which members of the public are asked to take shelter indoors as a safer option to evacuation, providing a level of protection from the impact or consequences of an emergency.
Shared responsibility	Shared responsibility refers to the collective obligations and accountabilities held by a range of actors. Definition derived from the SEMP <https://files.emv.vic.gov.au/2021-10/Victorian%20State%20Emergency%20Management%20Plan%20%28SEMP%29%20-%20Interactive%20and%20functional%20PDF%20document%20-%20October%202021.PDF?VersionId=C7TpaNlHZCZ9cA8xS0K0C.LuWfAhPJa3>
Stochastic effect (long-term health effects)	Effect resulting from damage in a single cell, such as cancer and heritable effects. Definition derived from the International Commission on Radiological Protection (ICPR) <http://icrpaedia.org/ICRP_Glossary>
Triage	The process by which casualties are sorted, prioritised and distributed, according to their need for first aid, resuscitation, emergency transportation and appropriate care.

Appendix C: Public information and community engagement arrangements

This appendix provides information on planning for and responding to a radiation emergency. It covers state and national governance arrangements, workforce planning, community insights, communications and engagement channels, and messaging development.

Governance

Victorian Government

Depending on the incident response level, the Department of Premier and Cabinet, DH, EMV and Victoria Police may need to work closely together to ensure consistent and coordinated messaging.

The Emergency Management Joint Public Information Committee (EMJPIC) coordinates whole of Victorian Government communications and provides strategic guidance for state-level communications and engagement planning and messaging.

Regional and Incident Joint Public Information Committees may also be established to help coordinate localised communications.

When an emergency occurs and affects the health and wellbeing of Victorians, DH will work closely with the State Control Centre and/or Regional and Incident Control Centres to provide real-time public information, warnings, and advice.

Australian Government

The Australian Government Department of Health and Aged Care coordinates national health messaging via the National Health Emergency Media Response Network.

Planning

Public information and community engagement planning

A state health radiation communications plan has been developed, and:

- outlines the unique state and national public information arrangements
- considers various scenarios, risks, and mitigations
- determines approaches for key communication functions including proactive and reactive media and social media, community and stakeholder engagement, advertising campaigns, strategic communications, and website creation and management.

Following a radiation emergency an incident specific communication plan should be developed based on the specific consequences.

It is essential that planning identifies targeted communications and engagement channels to reach people most at risk which may include, Aboriginal and Torres Strait Islanders, those from multicultural and multifaith communities, and vision and hearing-impaired Victorians.

It is also essential that arrangements are in place for the rapid translation of critical information across different channels.

Implementation

Public Information Unit

Upon notification of a radiation emergency and depending on the incident response level, the Department of Premier and Cabinet, the State Control Centre, Department of Health, and Victoria Police will work closely to issue warnings and advice and to coordinate communications and engagement. DH may embed Public Information Liaison Officers in the State Control Centre to facilitate and coordinate communications.

Other key functions will likely include media, digital and social media, strategic communications, community and stakeholder engagement, and an Information and Warnings Officer.

There may be a need to rapidly scale up a communications and engagement workforce and use surge staff from within DH or from across government.

Insights and community engagement

It may be important to draw on established community and industry networks to inform our communications and engagement response. DH can use established networks to test messaging and campaign materials to ensure it is effective and culturally and linguistically appropriate.

Daily media and social media monitoring may be required with insights and data reported to executives and key decision makers to inform the broader response.

Communications and engagement channels

In the event of a radiation emergency, DH will need to use a variety of communication and engagement channels including but not limited to media, paid advertising, digital and social media, as well as stakeholder and community engagement channels. The VicEmergency website, app, and telephone hotline along with the Emergency Alert telephony system may also be used to issue warnings and advice to impacted communities.

All messaging should be regularly reviewed and updated as the situation changes.

Media spokespeople and those delivering messaging should be trusted, confident and experienced subject matter experts authorised to speak on behalf of the Victorian Government.

Using community and faith leaders to develop and share messaging is also an effective strategy for reaching communities and consumers through trusted sources. Established local networks should also be used to help disseminate important or urgent messages to communities.

Messaging

Whole of Victorian Government messaging should be coordinated through the State Control Centre and reviewed and updated regularly.

Victorian Government departments and agencies are responsible for developing their own messaging about their response or impacts to their services; and managing appropriate departmental approvals.

See HESP section 6.2 for information on DH authorisation environment based on the incident response level.